Two-Lens Systems

SYNOPSIS

Every optical instrument worth owning has more than one lens in it, and not one of them needs an equation beyond the two you already have. You solve the first lens, take its image, and hand that image to the second lens as its object. That one sentence is the entire method.

BEFORE YOU START — PREDICT

Lens 1 inverts the object. Lens 2 then inverts what it is given. So is the final image upright or inverted compared with the original object? And what does that tell you about the sign of the *total* magnification?

PERFORMANCE EXPECTATIONS

1. **State** the rule that links the two stages of a multi-lens problem.
2. **Compute** the object distance for the second lens from the image distance of the first and the separation.
3. **Combine** the two magnifications into the magnification of the system.
4. **Recognise** when the intermediate image falls beyond the second lens, and say what its sign then means.

5. Two-lens system

THE RULE

The image of the first becomes the object of the second

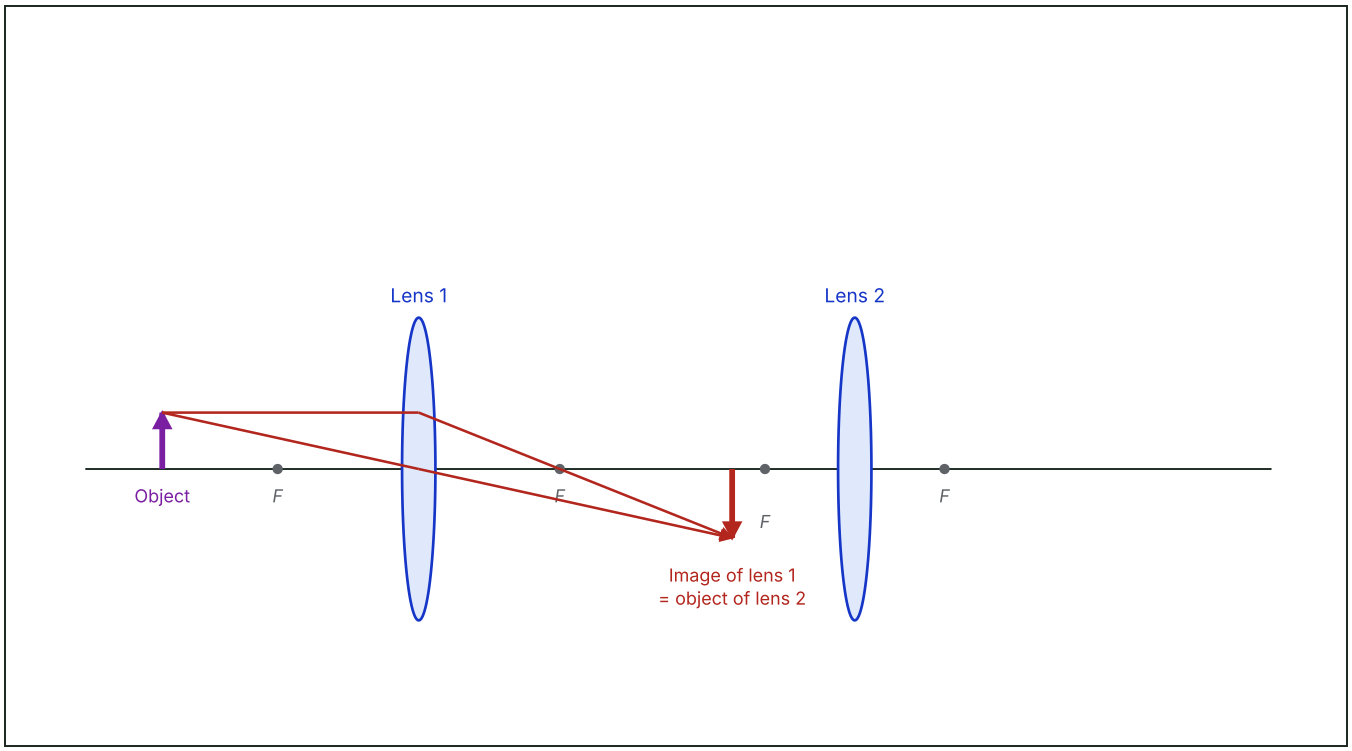


Figure 1. Two converging lenses, solved one at a time. Lens 1 forms a real inverted image between the lenses (red). That image is then treated as an ordinary object for lens 2 (purple) — nothing about the second calculation knows there was ever a first one.

Your job on Figure 1. Stage 1 is done: lens 1 has formed its image between the lenses. Now treat that image as an ordinary object and do stage 2 — trace two rays from *its* tip through lens 2 and mark the final image. Is it upright or inverted compared with the original object?

Work the lenses strictly in order. For lens 1 use the object you were given and find d_{i1}. Then, if the lenses are a distance L apart,

$$d_{o2} = L - d_{i1}$$

and feed that into the same thin lens equation for lens 2. The magnification of the whole system is the product

$$m = m_1 \times m_2$$

Watch the sign of d_{o2}. If lens 1 would have formed its image *past* lens 2, then $d_{i1} > L$ and d_{o2} comes out **negative**. That is the *virtual object* of the table in Note 8 — not an error, and the equation handles it without any special case.

Worked example — carry the construction through

1. In Figure 1, lens 1 has already formed its image between the two lenses. **Carry the construction through lens 2 by hand.** What do you use as the object, which rays do you draw, and what happens to the magnification?

THE PUNCHLINE

$$d_{o2} = L - d_{i1} \qquad m = m_1 m_2$$

Back to the prediction: two inversions give an **upright** final image, and the two negative magnifications multiply to a positive one. No new physics has entered this module since Note 6 — only the discipline of doing one lens at a time.

Looking ahead

That closes geometric optics: light treated strictly as rays that travel in straight lines and bend at surfaces. From **Note 10** the ray model runs out. Two beams of light will be made to arrive at the same place and produce *darkness*, which no amount of ray tracing can account for.

2. Two lenses are 40 cm apart and lens 1 forms its image 25 cm beyond itself. What is d_{o2}, in cm?

cm

3. $m_1 = -2$ and $m_2 = -3$. What is the magnification of the system?

4. Lens 1 would have formed its image 60 cm beyond itself, but lens 2 sits only 45 cm away. What does that make the object for lens 2?

○ An impossible arrangement; the problem has no solution. ○ A virtual object, with $d_{o2} = -15$ cm. ○ A real object, with $d_{o2} = +15$ cm. ○ A real object at the focal point of lens 2.

Check Clear

(adds about 420 px to the page; re-measure and re-declare the wrapper height) --> Recording to come. Print the guided notes (link just below) and fill them in as it goes.